Example 3.59 If the three angles in a triangle are in the ratio $1 : 2 : 3$, then prove that the corresponding sides are in the ratio $1 : \sqrt{3} : 2$.

Solution:
Let the angles be $\theta, 2\theta, 3\theta$.

Then $\theta + 2\theta + 3\theta = 180^\circ$

Thus, $\theta = 30^\circ$

Using the sine formula, $\dfrac{a}{\sin A} = \dfrac{b}{\sin B} = \dfrac{c}{\sin C}$, we have,

$$\frac{a}{\sin 30^\circ} = \frac{b}{\sin 60^\circ} = \frac{c}{\sin 90^\circ}$$

$$a : b : c = \sin 30^\circ : \sin 60^\circ : \sin 90^\circ$$

$$= \frac{1}{2} : \frac{\sqrt{3}}{2} : 1 = 1 : \sqrt{3} : 2$$

Example 3.60 In a $\triangle ABC$, prove that

$$(b + c)\cos A + (c + a)\cos B + (a + b)\cos C = a + b + c$$

Solution:

$$\begin{aligned} \text{L.H.S} &= b\cos A + c\cos A + c\cos B + a\cos B + a\cos C + b\cos C \\ &= b\cos C + c\cos B + c\cos A + a\cos C + b\cos A + a\cos B \\ &= a + b + c \quad \text{[by projection formula]} \end{aligned}$$

Example 3.61 In a triangle ABC, prove that $\dfrac{a^2 + b^2}{a^2 + c^2} = \dfrac{1 + \cos(A - B)\cos C}{1 + \cos(A - C)\cos B}$

Solution:
The sine formula is, $\dfrac{a}{\sin A} = \dfrac{b}{\sin B} = \dfrac{c}{\sin C} = 2R$

$$\begin{aligned} \text{L.H.S} &= \frac{a^2 + b^2}{a^2 + c^2} = \frac{(2R\sin A)^2 + (2R\sin B)^2}{(2R\sin A)^2 + (2R\sin C)^2} \\ &= \frac{\sin^2 A + \sin^2 B}{\sin^2 A + \sin^2 C} = \frac{1 - \cos^2 A + \sin^2 B}{1 - \cos^2 A + \sin^2 C} \\ &= \frac{1 - \left(\cos^2 A - \sin^2 B\right)}{1 - \left(\cos^2 A - \sin^2 C\right)} = \frac{1 - \cos(A + B)\cos(A - B)}{1 - \cos(A + C)\cos(A - C)} \\ &= \frac{1 + \cos(A - B)\cos C}{1 + \cos(A - C)\cos B}. \end{aligned}$$